

Datascience

1. Python basics

2. Numpy

- Numpy arrays
- Indexing and slicing on array
- Array operations

3. Pandas

- Data series
- Data frames
- Data cleaning
- Data aggregation
- Feature engineering
- Time series data
- Parallel processing

4. Visualization

- Matplotlib
- Seaborn
- Geographical plots

5. Statistics

- Descriptive statistics
- Sampling
- Frequency distributions
- Mean, Mode, Median
- Variance, Standard deviation, Bessel's correction
- Z-scores

6. Probability theory

- Estimating probability
- Addition rule

- Permutation and combinations
- Bayes Theorem

7.Hypothesis testing

8.Machine learning

- Overview about supervised and unsupervised ML
- Error metrics
- Linear regression
- Logistic regression
- SVM
- Decision tree
- Random forest
- K-nearest neighbors

9.Natural language processing

10.Neural networks and Deep learning

- Neural Networks
- Forward propagation
- Activation functions
- Loss functions
- Back propagation
- Tensorflow
- keras

11.Case studies

12.Clustering

- K-means clustering
- Mean-shift clustering

13.Case studies

14.Convolution neural networks

15.Big data

- Hadoop,MapReduce and spark
- Pyspark
- Case study on pyspark